UNIVERSITY OF MARYLAND

Laboratory for Control and Information Systems

Deep Learning-Enabled Diagnosis of Abdominal Aortic Aneurysm Using Pulse Volume Recording Waveforms: An In-Silico Study

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1. Motivation and Background

❖ Motivation

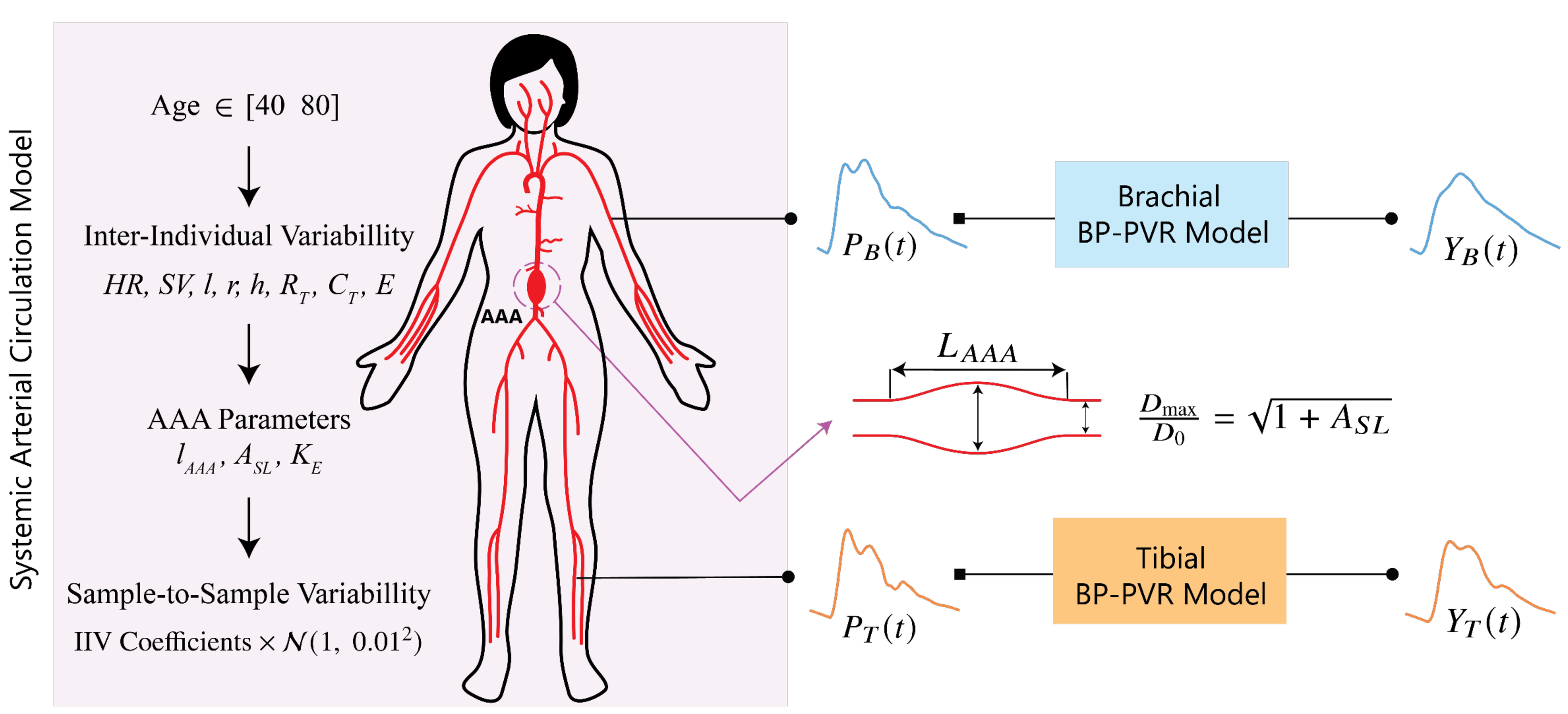
- To diagnose abdominal aortic aneurysm **(AAA)** with routine cuff-based measurements

❖ Background

- AAA is a localized dilation in the abdominal aorta.
- Definitive diagnosis of AAAs hinge on imaging methods, including ultrasound, CT, and MRI.
- More than 60% of privately insured patients in the US do not receive the recommended screening.
- Barriers include limited awareness of AAA, inconvenience of in-person appointments, and costs.
- AAA alters aortic geometry and compliance and leaves measurable imprints on arterial blood pressure **(BP)** waveforms.

2. Methods

❖ A 1D systemic arterial circulation + 0D BP-PVR model is used to generate a AAA synthetic cohort patient.



❖ AAA parameters

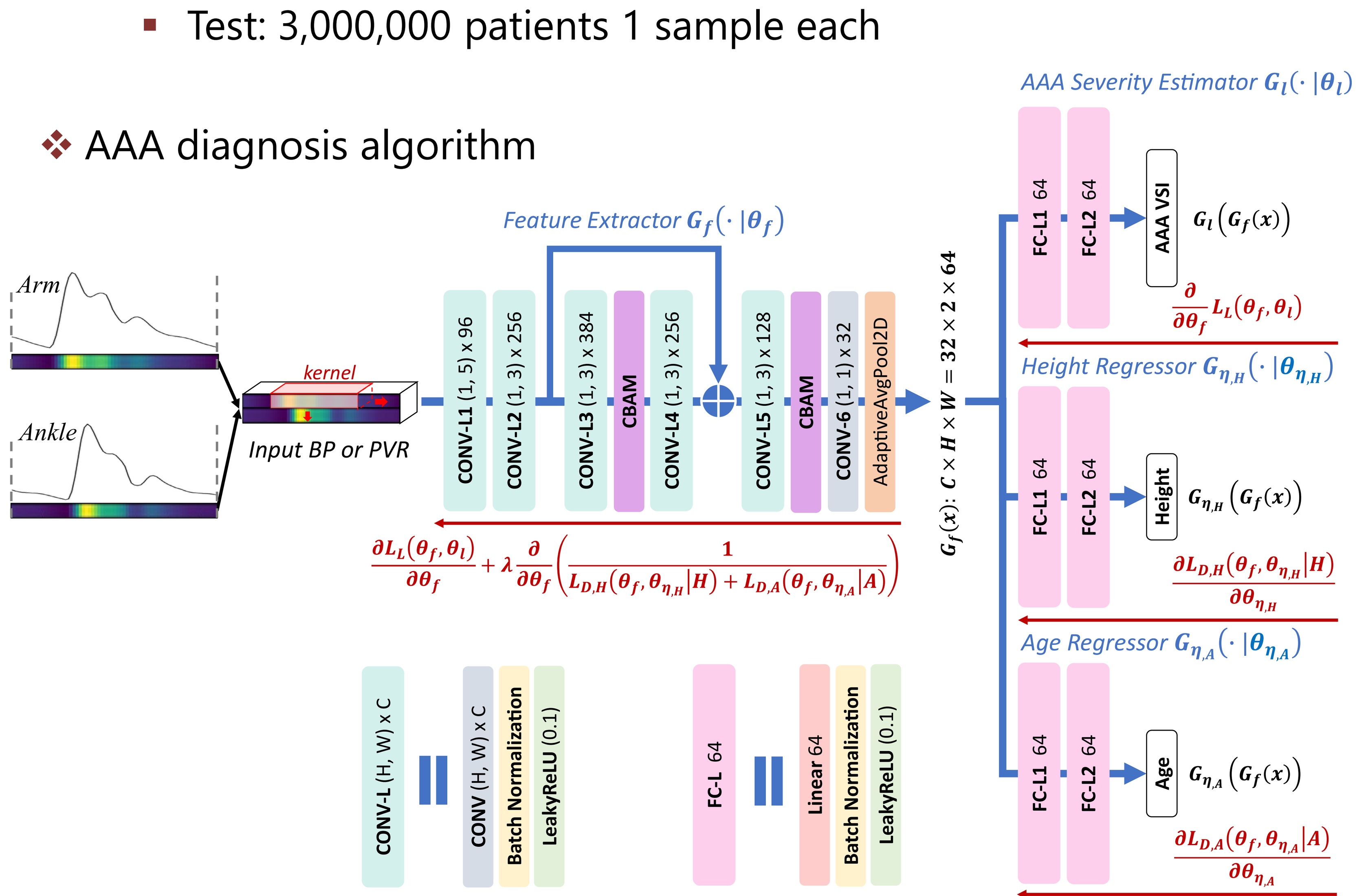
- A_{SL}: 0 – 5.25 → AAA maximum diameter
- L_{AAA}: 0.4 – 1 → AAA length
- k_E: 1.25 – 2.45 → AAA wall Young's modulus

$L_{AAA} = 1$ and $A_{SL} = 5.25$ → a 2.5-fold increase in AAA maximum diameter and ~3.6-fold increase in volume

❖ Generated Dataset

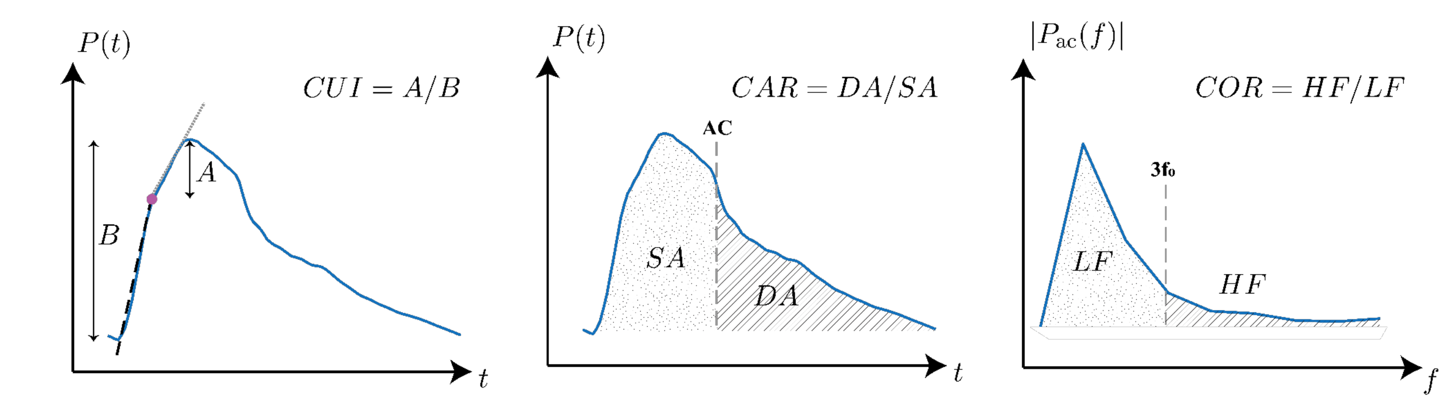
- Train:
  - Labeled: 200 patients 100 samples each
  - Unlabeled: 200 patients 100 samples each
- Validation: 200 patients 100 samples each
- Test: 3,000,000 patients 1 sample each

❖ AAA diagnosis algorithm



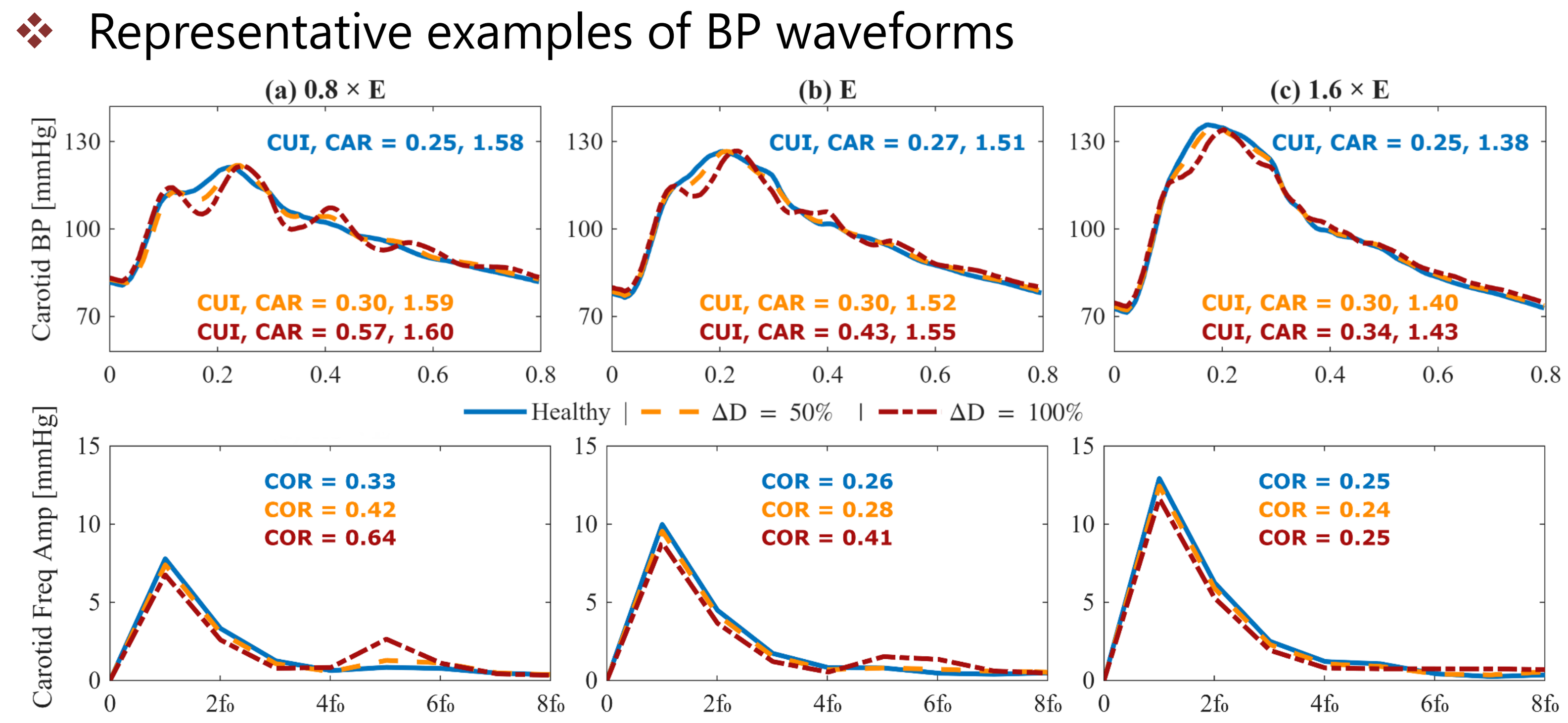
$$\frac{\partial L_I(\theta_f,\theta_I)}{\partial \theta_f} + \lambda \frac{\partial}{\partial \theta_f}\left(\frac{1}{L_{D,H}(\theta_f,\theta_{\eta,H}|H) + L_{D,A}(\theta_f,\theta_{\eta,A}|A)}\right)$$

❖ AAA-focused BP waveform analysis

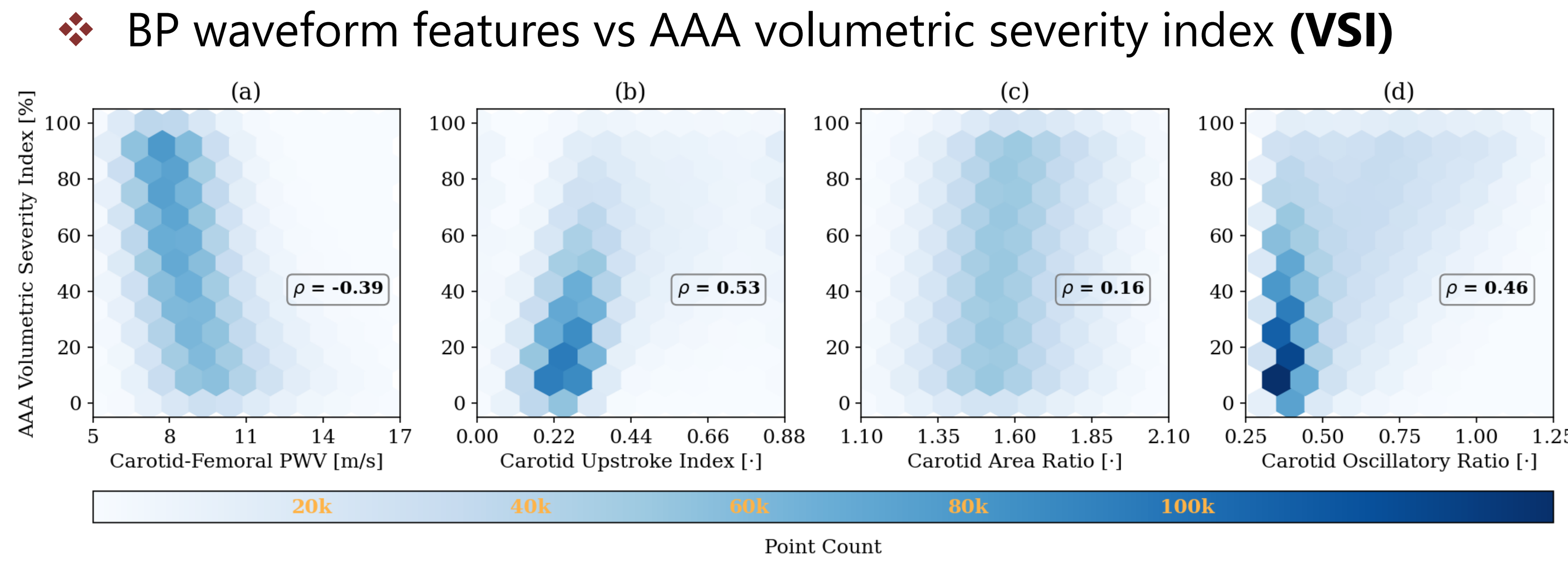


3. Results

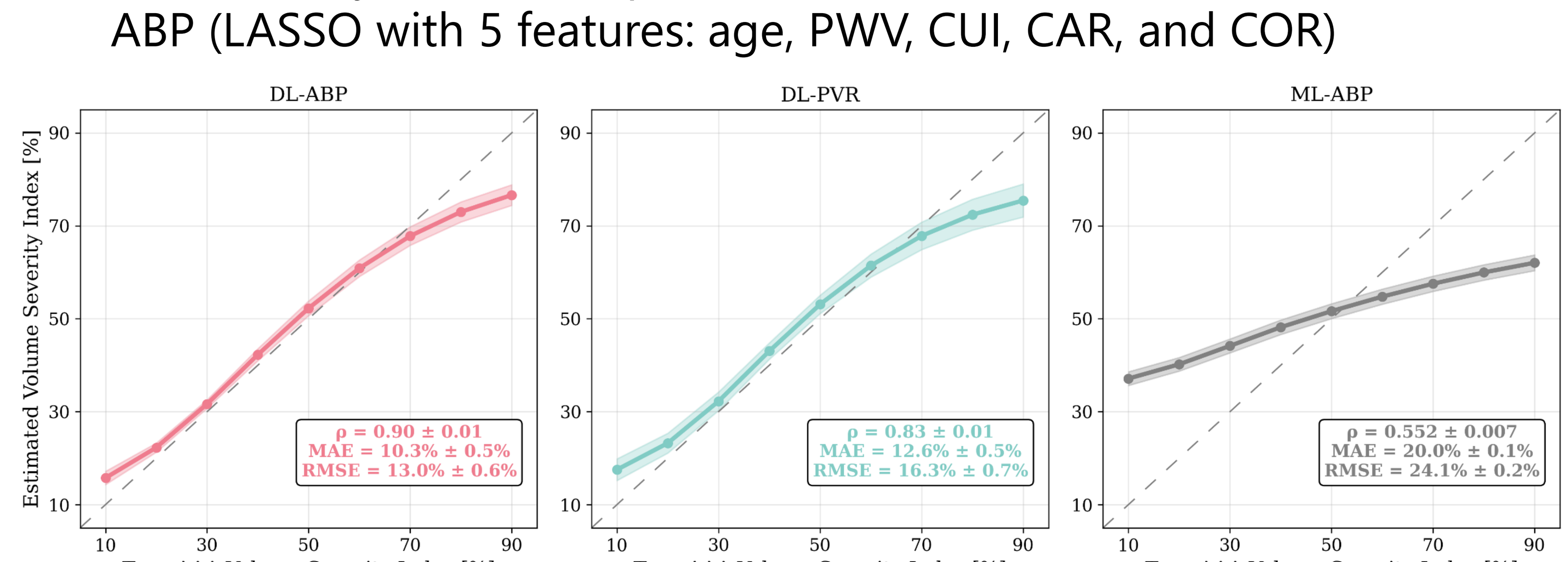
❖ Representative examples of BP waveforms



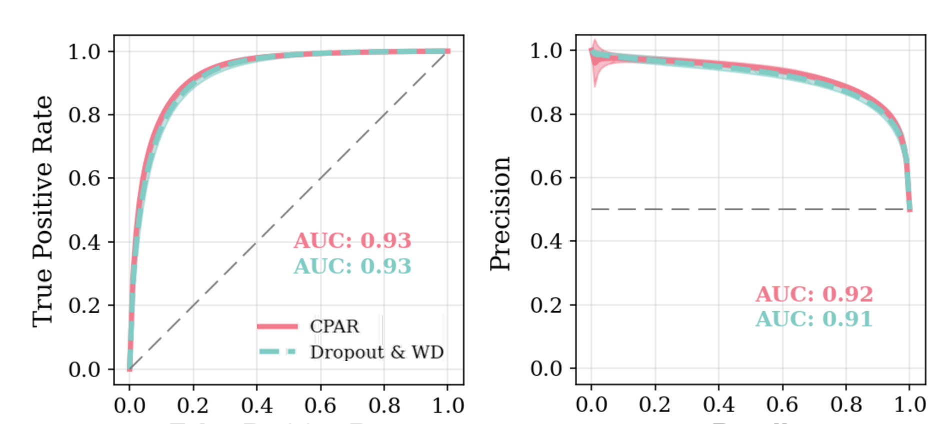
❖ BP waveform features vs AAA volumetric severity index **(VSI)**



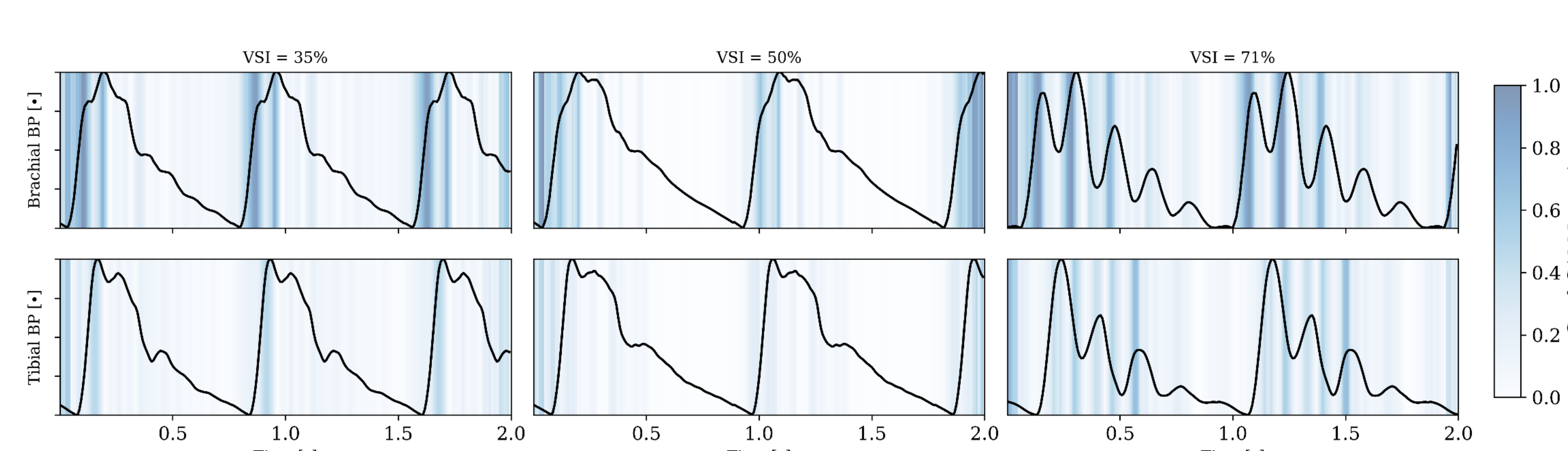
❖ AAA severity estimation performance of DL-ABP vs. DL-PVR vs. ML-ABP (LASSO with 5 features: age, PWV, CUI, CAR, and COR)



❖ Performance of DL-PVR with CPAR vs. dropout and weight decay



❖ Grad-CAM visualization of model attention for DL-ABP



Acknowledgment

Research supported by **the National Institutes of Health** under Grant 1R03EB032793-01 and **University of Maryland Graduate School Faculty-Student Research Award**.